

Hardware

Duke OLLI Digital Explorers SIG

David Shamlin, July 2026

Learning Objectives

- Knowledge
 - What is “hardware” and what are the kinds/types of hardware
 - Basic anatomy
 - Computation essentials: Processor + Memory + Storage
 - Interactivity: how you “talk to” a computer and how it “talks back”
 - Connectivity: to additional storage, to interactivity peripherals, and to the Internet
- Skills: None

Definition and categories

- In the digital domain, the term “hardware” refers to the things you can physically touch
- Primary types/kinds of hardware
 - Computers (“general purpose”): desktops, laptops, phones, and tablets
 - “Smart devices”: TVs, thermostats, doorbell cameras, digital assistants (e.g., Alexa, Google Home, etc)
 - Peripherals: mice, external keyboards, external drives, printers, styluses, etc
- Two key design principles that influence how we interact with computers:
 - Multi-user (desktops & laptops) vs single-user (a.k.a. “mobile devices”; phones and tablets)
 - Screen size/type: “classic” vs touch-screen

Categorization by brand

Computers can also be categorized by the business/organization that produces them. Most people today have multiple devices (e.g., desktop/laptop, phone, and/or tablet); having a highly unified/integrated ecosystem is a primary reason people use devices that are all from the same manufacturer.

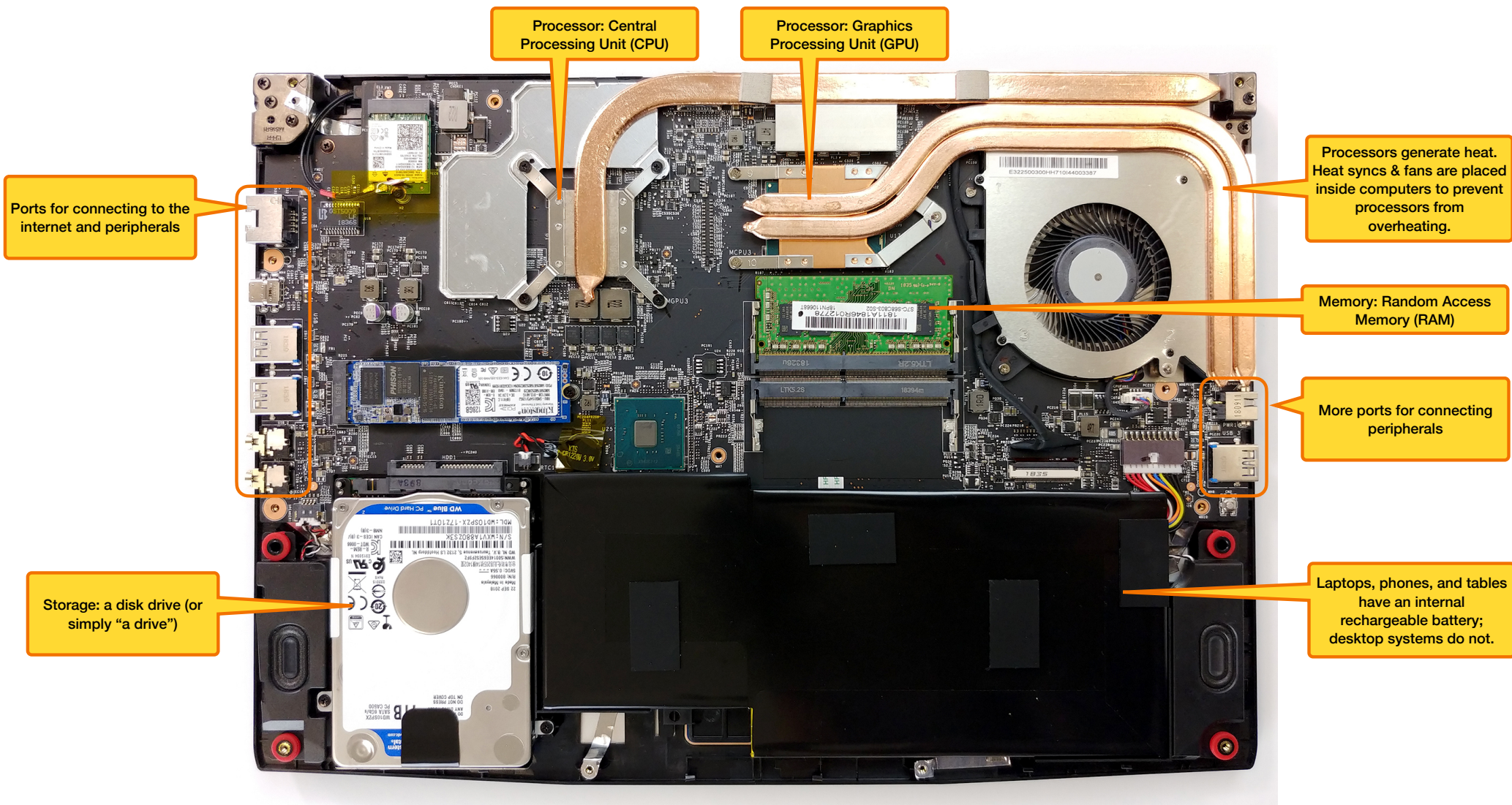
Apple is the only company that provides a complete end-to-end ecosystem. Google is a close second in that it produces operating systems for desktops/laptops, phones *and* tablets. However unlike Apple, you can purchase hardware for this category from other manufacturers besides Google; consequently, you have more options related to the monetary cost you pay for this class of computers.

Manufacturer	Operating System			
	Desktop/Laptop	Phone	Tablet	
Apple	macOS	iOS	iPadOS	• Highly regarded due to high value placed on quality, design aesthetic, and privacy/security • Design philosophy favors graphic design constructs
Microsoft	Windows		Windows	• Most popular computer OS
Google	ChromeOS	Android	Android	• ChromeOS is designed to favor/heavily depend on Google's consumer cloud services • Most smart devices use Android
	Linux			• The least common for personal/consumer use • People chose Linux when they want maximum customization and independence from the Tech Giants; there are many varieties of Linux • Most people would likely consider Linux to have the steepest/longest learning curve and to require more maintenance effort than the other brands/manufacturers listed

Computer Internals: What is inside a computer?

Computers are comprised of *many* constituent parts. (See the next page for an image of the inside of a laptop.) While the internal workings of a computer definitely has a high degree of complexity, the bits and pieces inside a computer can be grouped into three primary areas:

- Computation: the parts that enable computers to...well, compute: processor(s), memory, and storage
- Interactivity: the parts that enable you to “tell” the computer what to do (e.g., keyboard, mouse, trackpad, touchscreen) and the computer to “tell” you the results of what it did (e.g., screen, speakers)
- Connectivity: the parts that enable you to “connect” the computer to other devices (e.g., keyboard/mouse/drives, printers, monitors)



Computational components

The word “computer” originated in the 1640s and was used to refer a person whose occupation was to calculate—i.e., do math. The meaning of the word “computer” was expanded to include “calculating machine” in the 1940s. The [American Heritage Dictionary](#) defines *computer* as “a device that computes, especially a programmable electronic machine that performs high-speed mathematical or logical operations or that assembles, stores, correlates, or otherwise processes information; and defines *compute* as “to determine by mathematics, especially by numerical methods; to determine by the use of a computer.”

Computers are devices for capturing, storing, manipulating, and exchanging information. The information is mostly in the form of text, numbers, images (both still and moving), and sound. In the digital domain, information is referred to as “data” — sometimes as “digital data” or “binary data”.

There are three components found inside every computer are the primary components that enable them to perform these computations on data:

- **Processor(s):** handles the mathematical/logical operations referred to in the American Heritage Dictionary definition
- **Memory:** holds the data the processor(s) need(s) to perform the computations underway at a moment in time. Any/all data the processor(s) need in order to do computations requested by the user (i.e., you) resides in memory while the processor(s) complete those computations; when the processor is done with the computations that depend on the data being held in memory, that data is removed/deleted/purged from memory.
- **Storage:** holds data for the long term. When data that needs to be saved/retained for periods of time when it is not currently/actively being used by the processor, the data is held by the storage componentry.

More on Memory

- Memory is the holding place for the instructions and data a processor needs to access quickly. I.e., memory is where information is stored for immediate use by the processor. Memory is one of the basic functions of a computer, because without memory, a computer wouldn't be able to function.
- There are technically two types of computer memory: primary and secondary. When computer manufacturers state how much memory a device has, they are referring to primary memory; when they use the word “storage” they are referring to secondary memory. I.e., drives are secondary memory.
- The preferred “geek” term for primary memory is “random access memory” and abbreviated as “RAM”. The terms “random access memory” and “RAM” are used as frequently as “memory” in “geek-speak”.
- The image in the upper right hand corner of this page is a RAM stick. The memory in you computer looks similar. (Exactly what your computer looks like will depend on the amount of memory in your computer and the company that manufactured it.)

More on Storage

- From technopedia.com:
 - *Storage is a process through which digital data is saved within a data storage device by means of computing technology. Storage is a mechanism that enables a computer to retain data, either temporarily or permanently.*
 - *Storage saves digital data on a device, allowing it to be kept temporarily or permanently.*
 - *Storage devices include hard drives, flash drives, and cloud systems managed by systems like NAS, SAN, and DAS.*
 - *RAM [i.e., memory] is used for short-term tasks and is cleared when turned off, while storage keeps data permanently.*

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More on Storage: Types of storage devices

- When information/data is placed in storage, it is saved/written to a “disk drive” —or simply a “drive”. The terms “storage”, “drive”, and “disk drive” are used as synonyms.
- Drive technologies
 - Hard disk drive (HDD): an HDD is a mechanical device; i.e., it contains moving parts. Inside a HDD is a “platter” that physically spins and an “arm” that moves across the “platter” when data is read from/ written to the drive. (Think of a vinyl record player; the “platter” is like the vinyl record and the record player’s arm is like the drive’s arm.)
 - Solid-state disk drive (SSD): an SSD is a solid-state device; i.e., it contains **no** moving parts. Instead integrated circuits are used to store data.
 - SSD are superior to HDD for two reasons:
 - SSDs are faster than HDDs
Although the difference in speeds between SSD and HDD won’t often be noticeable to us due to the types of things we do with our devices.
 - SSD are less likely to break than HDDs
Given that HDDs are mechanical devices, they can “wear out”; dropping/jarring them can also damage them.
 - SSDs are more expensive than HDDs.
 - Most computers sold today contain SSD drives.
- Drive are encased in different form factors (i.e., with different physical and shapes)
 - Internal: All computers have a drive inside them; these “internal drives” are physically constructed in ways to be optimally conducive to residing inside computers
 - External: You can purchase drives that you can attach to your computer to increase the amount of storage available to your device.
 - External drive: Essentially, the same as an internal drive but inside a more protective casing than an internal drive and with a port use to connect the external drive to your computer (typically) with a USB cable. Some external drives draw the power they need to function from the computer you connect them to; others must be plugged into an electrical outlet.
 - Thumb drive / USB drive / flash drive: SSDs inside a very small casing.
 - Memory card / flash cards: Essentially the same as a thumb drive but in an even smaller casing.
- Cloud drive: From the point of view of a user of a digital device, an argument can be made that cloud drives are not hardware; regardless...a cloud drive is essentially an external drive owned and maintained by a company that provides cloud services. From our point of view as consumer users, a cloud drive is a **virtual** device (which is just a fancy way of saying “using software to emulate a physical device”). We cannot physically touch it whereas we can physically touch the other types of external drives listed above. Cloud drives depend on the computer from which you use it being connected to the Internet.

More on Storage: Examples of drives

Internal HDD



The image on the left is a fully encased internal HDD. The image on the right is an internal HDD with its top protective covering removed. The platter and arm are highlighted with yellow arrows in the image on the right.

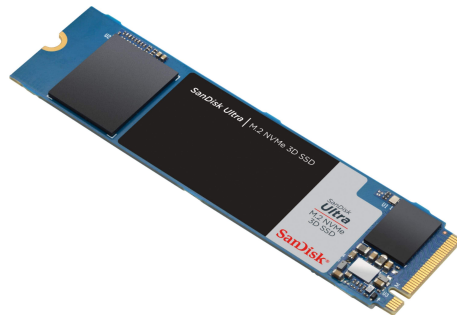
External Drive



The image on the left is of an external drive that is about the size of a 300 page hardback book. The image on the right is on an external drive that is about the size of a pack of playing cards. External drives can be either HDD or SSD. The image on the left is likely an HDD; the image on the right is an SSD. The cable attached to the drive on the right is a USB cable that you would plug into a USB port on your computer. The drive on the right has a USB port on its back side (not visible in this image) and you would plug one end of a USB cable into that port on external drive and the other end of the cable into a USB port on your computer; the external drive on the right uses a power cord that runs from the back of the external drive to a power outlet.

External drives that have to be plugged into a power outlet typically have on/off switches.

Internal SSD

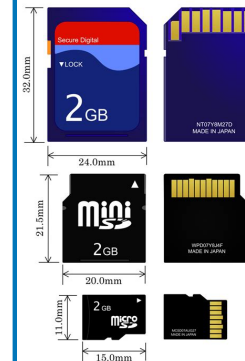


Thumb Drive



Thumb drives are SSDs (never HDDs) in a fairly small casing. The two shown here are about the size of a adult's thumb. The drive on the left has a USB-C type plug; the drive on the right has a USB-2/3 type plug.

Memory Card



Memory cards are popular with digital photographers and videographers as most digital cameras use memory cards; here memory cards play the role that photographic film and VHS tapes played in the past.

Memory cards come in 5-6 common physical sizes. The three common sizes are shown in the images on the right. The largest at the top is about the size of postage stamp; the smallest at the bottom is about the size of a fingernail.

More on Storage: When/why would I use an external drive?

The most common reasons people use an external drive are:

- Backups

“Backing up a computer” refers to the act of copying data stored on the computer's internal drive to an external drive. In the unfortunate event that the computer stops running or is lost/stolen, you still have your valuable data on the external drive.

- A need to increase the storage capacity of a computer

It's possible to “fill up” a drive. You can experience erratic/problematic behaviors with a computer when 90% or more of its internal drive storage space is utilized. If/when this occurs, and you are (1) not ready to replace your computer and/or (2) not willing/able to simply delete some of the files you have stored on your internal drive, you can move some of the files from your internal drive to an external drive to free up space on your internal drive.

People who use their computer to do things that involve large volumes of data (a photographer or videographer for example) may opt to use an external drive to store that data—leaving the internal drive free to store data for other types of work they do on their computer (that is not so data intensive). [I seriously doubt anyone in the group (besides myself) uses their device(s) in ways that involve this scale of data.]

- Transfer data from one computer to another

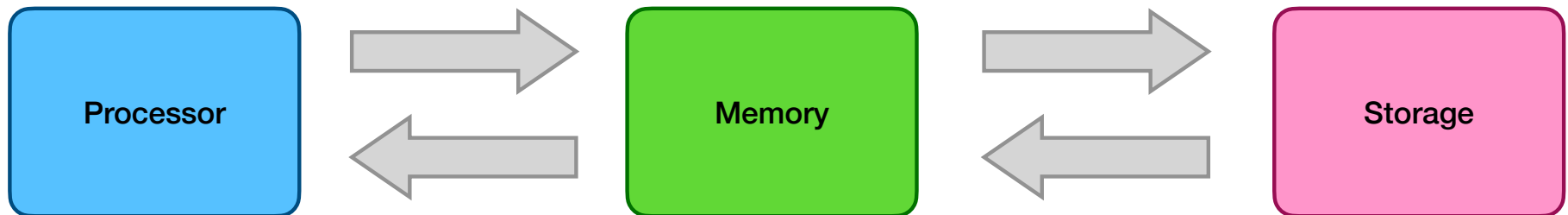
Sometimes you may want/need to copy files from one computer to another; when you want/need to copy more than “a handful” of files you want/need, an external drive can be convenient. While there are ways to send copies of files to another device over the internet, many of those ways are cumbersome when you want/need to copy more than approximately 10 files; in such situations, you may find it convenient to use an external drive.

More on Processors

- Computers contain two kinds of processors:
 - Central Processing Unit (CPU): manages system operations and executing instruction required to run software
 - Graphics Processing Unit (GPU): handles large-scale computations that can be performed in parallel, such as graphics and data processing (e.g., AI-type computations).
- A processor can perform a finite set of operations. Most processors in use today have in the neighborhood of 300-500 unique operations.
- These operations can be grouped into groups based on their purpose. E.g., there are operations for
 - Math (e.g., addition, subtraction, division, multiplication, etc)
 - Comparisons (e.g., $x = y$, $x < y$, $x \geq y$, etc)
 - Logic (specifically formal boolean logic: if condition is true, then do this else do that)

The relationship between processor, memory, and storage

- When executing an operation, the processor needs data to perform the operation on. Imagine the processor is attempting to compute the sum of two numbers; the processor needs “to know” the two numbers. The processor expects the data to be found in memory at the moment the processor is performing the operation.
- The data the processor needs to complete an operation is sent from memory to the processor. The processor then computes the result of the performing the operation on the data transferred to it from memory and produces the result of the operation. Again, imagine the processor is adding two numbers; the result of is the sum. The result of the operation is sent from the processor to memory.
- The data used to perform an operation is referred to as **input**. The data resulting from an operation is referred to as **output**. The concept of data acting as input and/or output is so central to digital systems that it is typically referred to using the acronym “IO” (short for “input/output”; sometimes written “I/O”; pronounced as “eye oh”).
- Data only resides in memory when the software that “owns” that data is running; when said software stops running, its data is deleted/purged/removed from memory.
- If/when the data in memory needs to be permanently save, it is transferred from memory to storage. This is done before any data is deleted from memory and the “owning” software stops running. (This is what happens when you save a document.)



Demo: Processor, memory, and storage

- Creating a new document with Pages
- Opening/editing an existing Pages document
- Creating/modifying a Contacts entry
- Calculating the sum of a set of numbers in a spreadsheet